

Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is a branch of computer science that focuses on building intelligent systems capable of performing tasks that normally require human intelligence. These tasks include problem solving, decision making, speech recognition, and language understanding.

Machine Learning (ML) is a subset of AI that enables systems to learn patterns from data and improve performance without being explicitly programmed. Machine learning algorithms are widely used in recommendation systems, image recognition, fraud detection, and predictive analytics.

Types of Machine Learning

Supervised Learning

Supervised learning uses labeled datasets to train models. Common algorithms include Linear Regression, Logistic Regression, Decision Trees, and Support Vector Machines.

Unsupervised Learning

Unsupervised learning works with unlabeled data and discovers hidden patterns. Clustering and dimensionality reduction are common applications.

Reinforcement Learning

Reinforcement learning trains agents through rewards and penalties. It is commonly used in robotics and gaming systems.

Deep Learning

Deep Learning is a specialized branch of machine learning based on artificial neural networks. Deep learning models can automatically learn hierarchical representations from large amounts of data.

Natural Language Processing

Natural Language Processing (NLP) enables machines to understand and generate human language. NLP applications include chatbots, machine translation, text summarization, and sentiment analysis.

Generative AI

Generative AI refers to models capable of generating text, images, code, audio, and videos. Large Language Models (LLMs) such as GPT are examples of Generative AI systems.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with language generation. In RAG systems, relevant documents are retrieved from a vector database and provided as context to the language model.

Agentic AI

Agentic AI systems use autonomous agents that can reason, plan, and use tools to complete tasks. Frameworks such as LangChain help developers build agent-based AI applications.